Aakash Gurung

github.com/GrgAakash   aakash.grg159@gmail.com
grgaakash.github.io   Tuscaloosa, AL

Education

Expected Dec 2026	M.A. and B.S. in Mathematics; Minor: Digital, Public, and Professional Writing, The University of Alabama, Tuscaloosa, AL

Research Interests

I am broadly interested in **Algebraic Combinatorics**, **Applied Mathematics**, and **Geometry**.

Research Experience

Present – June 2026
Multi-Agent AI for Mathematical Reasoning, RIPS, UCLA,
Academic Mentor: Mehmet Yigit Turali
Industry Sponsor: OpenAI
Industry Mentors: Giambattista Parascandolo (GB) and Professor Gregory Valiant
- Along with three fellow research interns, developing and evaluating multi-agent AI systems for mathematical reasoning and proof-oriented research workflows. Applying the system to a research problem under Professor Kyungyong Lee's guidance.

AI for Mathematics | Mathematical Reasoning | Multi-Agent Systems

Present – Feb 2026
Rectifiability, University of Alabama,
Mentor: Professor Simon Bortz
- Along with a postdoctoral researcher, studying rectifiability characterization in the parabolic setting and connections to Riesz transforms and tangent measures.

Geometric Measure Theory | Rectifiability

Present – Feb 2025
Combinatorial Perspective of (q,t)-Catalan Symmetry, University of Alabama,
Mentor: Professor Kyungyong Lee
- Working on a subset of the combinatorial proof of the (q,t)-Catalan symmetry.

Algebraic Combinatorics | Catalan Numbers

July 2026 – March 2026
LLM Distillation, University of Alabama,
Mentors: Professor Chuntian Wang and Professor Jia Zhao
- Along with a graduate student, worked on LLM distillation in network-based epidemic simulation.

LLM Distillation | Epidemic Modeling | Network-Based Simulation

Dec 2025 – May 2025
Finite-Size Effects in Epidemic Models, University of Alabama Mathematics Summer REU,
Advisors: Professors Chuntian Wang, Yuanyuan Song, Yuanzhen Shao
- Along with four fellow student researchers, worked on developing and analyzing the dynamics of an epidemic compartment model (SIHRS), incorporating vital dynamics through independent Poisson clocks and a martingale formulation. Used real-world county data from COVID-19 to show the model's usefulness. Also worked on the global stability analysis of the model.

Applied Mathematics | Epidemic Modeling | Stochastic Analysis

June 2024 – Feb 2024
Game of Cycles on Maximal Plane Graphs, CUNY Research Scholars Program,
Advisor: Professor Malgorzata Marciniak
- Defined "IO Maximal Plane Graphs" and analyzed invariant properties to determine the game outcome.
- Established that the winning strategy is determined by the parity of the graph's vertices.

Combinatorial Game Theory | Graph Theory

Aug 2023 – June 2023
Method of Brackets and Bessel Function Integrals, Polymath Jr 2023,
Advisor: Professor Victor H. Moll
- Applied the "Method of Brackets" to provide rigorous proofs for entries involving Bessel functions of the first and second kind from the Gradshteyn and Ryzhik tables.

Special Functions | Integral Calculus | Bessel Functions

Aug 2023 – Mar 2023
Continued Fractions, a-Fibonacci Numbers, and Middle b-Noise, Independent Project,
Advisor: Professor Cheng Han Pan
- Generalized palindromic continued fractions $[1,\ldots,1,3,1,\ldots,1]$ to $[a,\ldots,a,b,a,\ldots,a]$ using a-Fibonacci sequences.
- Showed that the a-th metallic ratio limit is invariant under the middle noise term b.

Number Theory | Continued Fractions | Fibonacci Sequences

Preprints

Submitted — **A. Gurung**, S. Wagle, A. Carr, C. McCann, K. Kodatt, Y. Song, Y. Shao, C. Wang. "An exploration of finite-size effects in the dynamics of epidemic compartmental modelling."

Expository/Other Works

2024 — **A. Gurung** and C.-H. Pan. "Continued Fractions, a-Fibonacci numbers, and the middle b-noise," *Mathematics Exchange*, 18(1), 77–87.

2024 — (with the Polymath Jr. Group). "The integrals in Gradshteyn and Ryzhik. Part 34: Bessel functions," *Scientia Series A: Mathematical Sciences*, 34, 109–129.

Conferences & Workshops

March 2026 — AMS Southeastern Sectional Meeting, **Presenter**

May 2024 — CUNY Undergraduate Research Day 2024, **Presenter**

Aug 2024 — MathFest 2024, **Presenter**

Sept–Nov 2024 — Preliminary Arizona Winter School 2024: Symmetries of Root Systems, Attendee

Honors & Awards

2026 — Thomas W. Palmer Math Award, University of Alabama

2026 — Randall Outstanding Undergraduate Research Award, University of Alabama

2025 — ASSURE Grant, University of Alabama

2024 — Best Poster Award, CUNY Undergraduate Research Day

2023 — Samuel J. Steinberger, Jr. Memorial Award, Juniata College

2020–2021 — Nepal Mathematical Olympiad (Top 12); Nepal Astronomy Olympiad (Rank 1); USA Astronomy & Astrophysics Competition (National Qualifier)

Work Experience

May 2026
Aug 2025

IT Service Desk Student Assistant, University of Alabama, Tuscaloosa, AL

- Provide timely software and technology support to resolve user issues efficiently.

IT Support | Technical Support

May 2025
Sept 2024

Peer Tutor, Mathematics Technology Learning Center, Tuscaloosa, AL

- Drop-in tutor for Calculus 1, 2, and 3.
- Run recitation classes for Calculus 1.

Calculus | Mathematics Education | Tutoring

Skills

Programming — Python (NumPy, Pandas, SciPy), Julia, MATLAB, JavaScript, HTML, CSS

Tools — PowerQuery, LaTeX, Git

Other — Grant Writing